Code No: 111AH **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year Examinations, June - 2022
ENGINEERING DRAWING
(Common to CSE, MIE, PTM)

Time: 3 Hours **Max. Marks: 75**

Answer any three questions
All questions carry equal marks
- - -

1.a) An area of 200 cm^2 on a map represents 50 km^2 of actual land area. Draw a diagonal scale to measure a distance up to a single decametre. Mark off the following distances on the scale: (i) 5 km, 7 hectometres and 6 decametres, (ii) 3.04 km and (iii) 458 decametres.

b) A semicircle of 50 mm diameter has a straight line AB of 80 mm length tangent to it at the end point, A of its diameter AC. Draw the paths of the end points A and B of the straight line AB, when AB rolls on the semicircle without slipping. Name the Curve. [12+13]

2.a) A straight line AB has its end A 15 mm above the HP and 10 mm in front of the VP. The other end B is 25 mm in front of the VP. The VT is 10 mm above the HP. Draw the projections of the line if the distance between end projectors is 25 mm and find its true length and true angles of inclination with the HP and the VP. Locate the HT.

b) Draw the projections of a triangular plate of 30 mm sides, with one of its sides AB in the VP and with its surface inclined at 60^0 to the VP. [13+12]

3. A cylinder of diameter 50 mm and length of the axis 65 mm has its axis parallel to the VP and inclined at 30^0 to the HP. It is cut by a cutting plane perpendicular to the HP, inclined at 30^0 to the VP, and passing through a point P on the axis 25 mm from the top end. Draw the top view, the sectional front view, and the true shape of the section of the cylinder. [25]

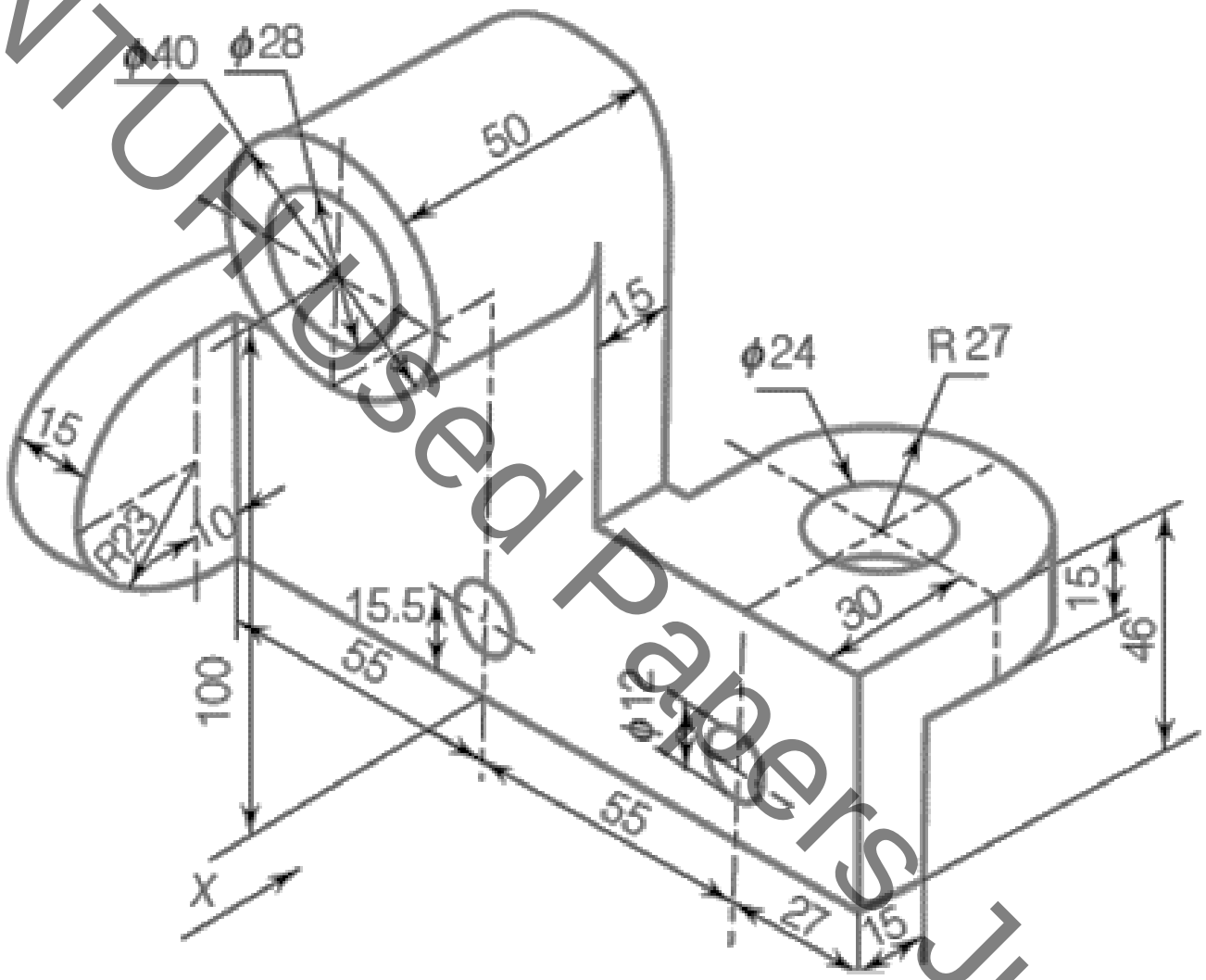
4. A cone of base diameter 80 mm and axis 100 mm, is resting on its base on the H.P. It is completely penetrated by a cylinder of base diameter 40 mm. The axes of the solids intersect each other at right angles, 30 mm above the base of the cone. Draw the projections of the combination and show curves of intersection. [25]

5. A cylinder of diameter 50 mm and length of the axis 65 mm rests on its base with the axis perpendicular to the HP. It is cut by the cutting plane perpendicular to the VP, inclined at 45^0 to the HP and passing through a point on axis 25 mm from the top. Draw the front view, the sectional top view, and the development of the lateral surface of the cylinder. [25]

JNTUH Used Papers June 22

(Contd…)

6. Pictorial view of an object is shown in figure. Using first angle projection, draw its (a) full sectional front view, (b) top view and (c) side view. (All dimensions are in mm). [25]



--ooOoo--

JNTUH Used Papers June 22